INCH-POUND

MIL-DTL-3970/13D
3 February 2009
SUPERSEDING
MIL-DTL-3970/13C
25 November 1998

DETAIL SPECIFICATION SHEET

WAVEGUIDE ASSEMBLIES, RIGID, 90-DEGREE H-PLANE BEND

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the waveguide assemblies described herein shall consist of this specification sheet and MIL-DTL-3970.

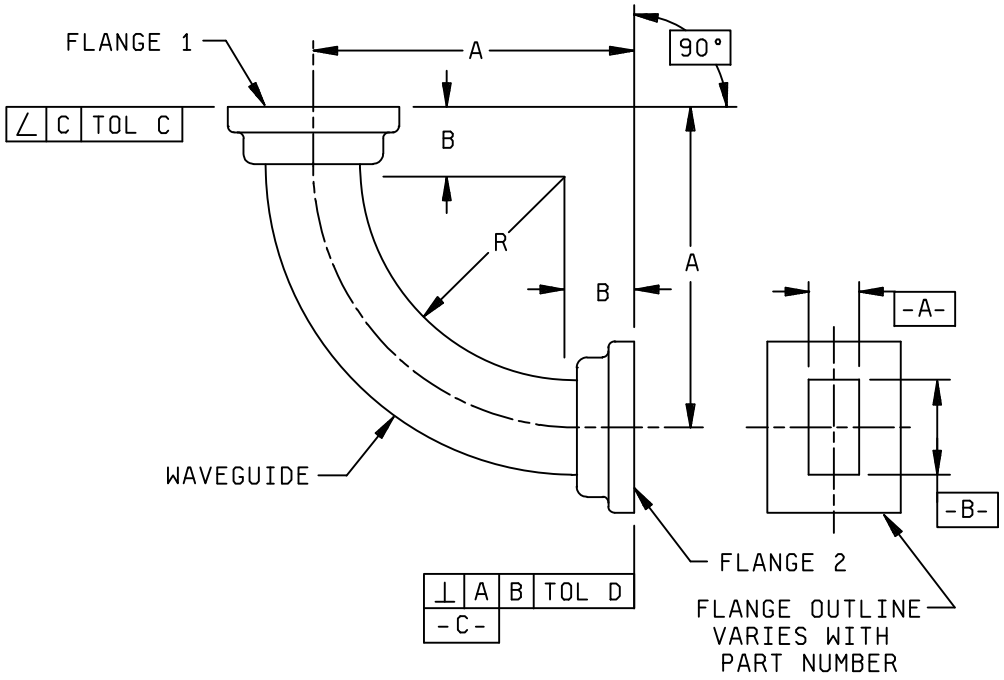

FIGURE 1. Assembly configuration.

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TABLE I. Dash number, electrical and physical characteristics, and dimensions.

Dash no.	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Pressurization lb_f/in^2 gage	Dimensions 1/ 2/ 3/ A	B nom	C	D	R
				GHz	dB							
001	1-043	56-001	61-002	2.6 - 3.95	---	1.05	15	10.500 (266.70) ±.030 (±0.76)	1.500 (38.10)	.115 (2.92)	.012 (0.30)	7.500 (190.50) ±.125 (±3.18)
002	1-055	57-002	62-002	3.95 - 5.85	---	1.05	30	7.000 (177.80) ±.030 (±0.76)	1.000 (25.40)	.077 (1.96)	.008 (0.20)	5.000 (127.00) ±.125 (±3.18)
003	1-067	55-001	60-001	5.85 - 8.20	---	1.03	30	4.875 (123.83) ±.025 (±0.64)	.750 (19.05)	.056 (1.42)	.006 (0.15)	3.375 (85.72) ±.125 (±3.18)
004	1-073	53-002	59-007	7.05 - 10.0	---	1.07	30	4.062 (103.17) ±.020 (±0.51)	.687 (17.45)	.044 (1.12)	.005 (0.13)	2.750 (69.85) ±.063 (±1.60)
005	1-079	53-001	59-006	8.20 - 12.4	---	1.05	30	3.562 (90.47) ±.020 (±0.51)	.562 (14.27)	.038 (0.97)	.004 (0.10)	2.500 (63.50) ±.063 (±1.60)
008	1-077	53-003	59-008	8.2 - 12.4	---	1.05	30	3.562 (90.47) ±.015 (±0.38)	.562 (14.27)	.038 (0.97)	.004 (0.10)	2.500 (63.50) ±.063 (±1.60)
009	1-071	53-004	59-009	7.05 - 10.0	---	1.07	30	4.062 (103.17) ±.020 (±0.51)	.687 (17.45)	.044 (1.12)	.005 (0.13)	2.750 (69.85) ±.063 (±1.60)
010	1-041	56-002	61-001	2.60 - 3.95	---	1.05	15	10.500 (266.70) ±.030 (±0.76)	1.500 (38.10)	.115 (2.92)	.012 (0.30)	7.500 (190.50) ±.125 (±3.18)
011	1-089	53-005	59-001	12.4 - 18.0	.07	1.04	30	2.601 (66.07) ±.015 (±0.38)	.500 (12.70)	.028 (0.71)	.003 (0.08)	1.750 (44.45) ±.063 (±1.60)
012	1-053	57-001	62-001	3.95 - 5.85	---	1.05	30	7.000 (177.80) ±.030 (±0.76)	1.000 (25.40)	.076 (1.93)	.008 (0.20)	5.000 (127.00) ±.125 (±3.18)
014	1-065	55-002	60-002	5.85 - 8.20	---	1.03	30	4.875 (123.82) ±.025 (±0.64)	.750 (19.05)	.056 (1.42)	.006 (0.15)	3.375 (85.72) ±.125 (±3.18)

See footnotes at end of table.

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TABLE I. Dash number, electrical and physical characteristics, and dimensions - Continued.

Dash no.	Waveguide M85/	Flange 1 M3922/	Flange 2 M3922/	Frequency range TE_{10} mode	Insertion loss max	VSWR max	Pressurization lb_f/in^2 gage	Dimensions 1/ 2/ 3/ A	B nom	C	D	R
				GHz	dB							
015	1-093	53-005	59-001	12.4 - 18.0	.07	1.04	30	2.601 (66.07) ±.015 (±0.38)	.500 (12.70)	.028 (0.71)	.003 (0.08)	1.750 (44.45) ±.063 (±1.60)
017	1-018	52-002	52-002	1.12 - 1.70	---	1.05	30	7.810 (198.37) ±.010 (±0.25)	1.116 (28.35)	.106 (2.69)	.028 (0.71)	3.364 (85.45) ±.125 (±3.18)
031	1-018	52-002	52-002	1.12 - 1.70	.10	1.05	1.5	14.000 (355.60) ±.015 (±0.38)	2.000 (50.80)	.160 (4.06)	.029 (0.74)	8.670 (220.22) ±.015 (±0.38)
032	1-029	52-006	52-006	1.70 - 2.60	.10	1.05	3.0	10.000 (254.00) ±.015 (±0.38)	2.000 (50.80)	.101 (2.57)	.019 (0.48)	5.770 (146.56) ±.015 (±0.38)

1/ Dimensions are in inches.
2/ Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3/ Metric equivalents are in parentheses.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Electrical characteristics: See table I.

Part or Identifying Number (PIN): M3970/13-(dash number from table I).

Cross reference: See table II.

TABLE II. Cross reference.

PIN M3970/13-	Type designation	PIN M3970/13-	Type designation
001	UG-712/U	009	UG-720/U
002	UG-713/U	010	UG-721/U
003	UG-714/U	011	UG-722/U
004	UG-715/U	012	UG-723/U
005	UG-716/U	013	UG-724/U
006	UG-717/U	014	UG-725/U
007	UG-718/U	015	UG-726/U
008	UG-719/U	016	UG-727/U

NOTE: Assemblies covered by this specification are substitutable for the following commercial types:

PIN M3970/13-	Manufacturer CAGE code	Commercial type
017	10001	2938971-1
017	01456	650BH49
017	05865	L2257
017	17250	L11114
031	52564	3705001
032	52564	4005001

Referenced documents. This specification sheet only references MIL-DTL-3970.

Marginal notations are not used on this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Preparing activity:
DLA - CC

(Project 5985-2009-008)

Review activities:
Army - MI
Navy - AS, CG, MC, OS
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil/

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